

DATA 890 Final Project

Unsupervised Quantification of Social Behavior

Riley Harper



Outline

Introduction

- Pose Estimation
- SLEAP

Motivations

- Neurobiology
- MOUSE_TRAP

Methods

- Dynamic Graphs

Results

- Keypoint-MoSeq



Section Overview

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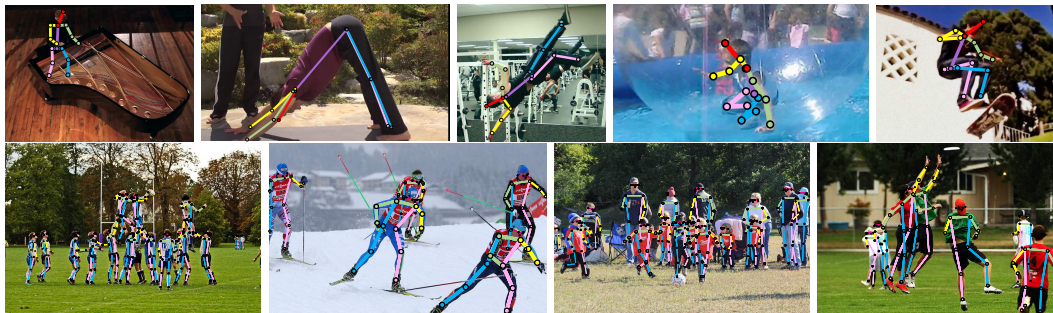


Objective

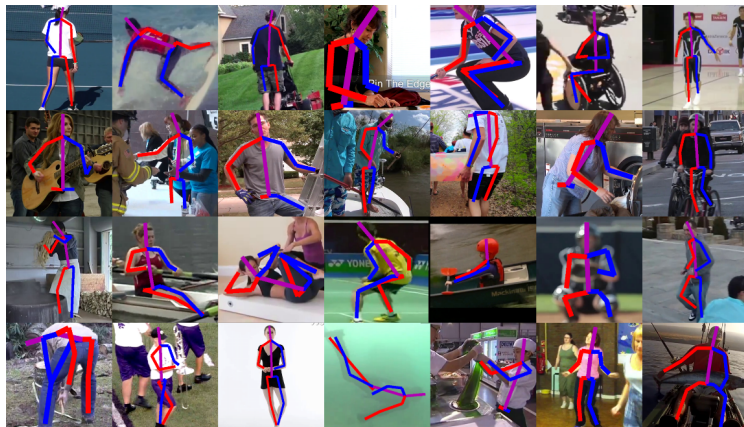
Given an image or video frame, automatically locate a set of predefined body keypoints for each animal, producing a graph representation of pose.



Example from Sun et al. 2019 (6K+ citations)

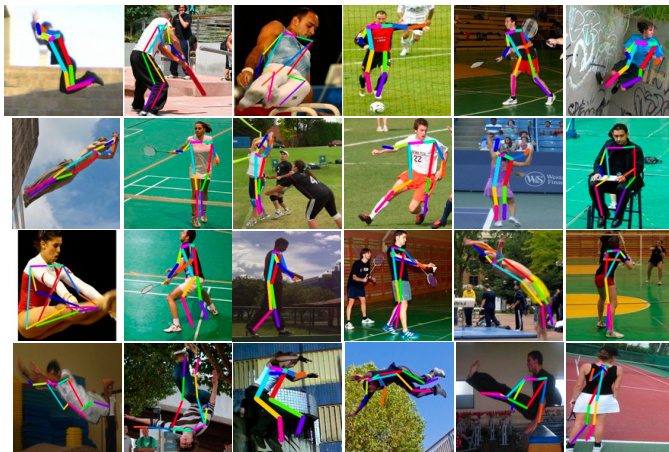


Example from Newell, Yang, and Deng 2016 (7.1K+ citations)



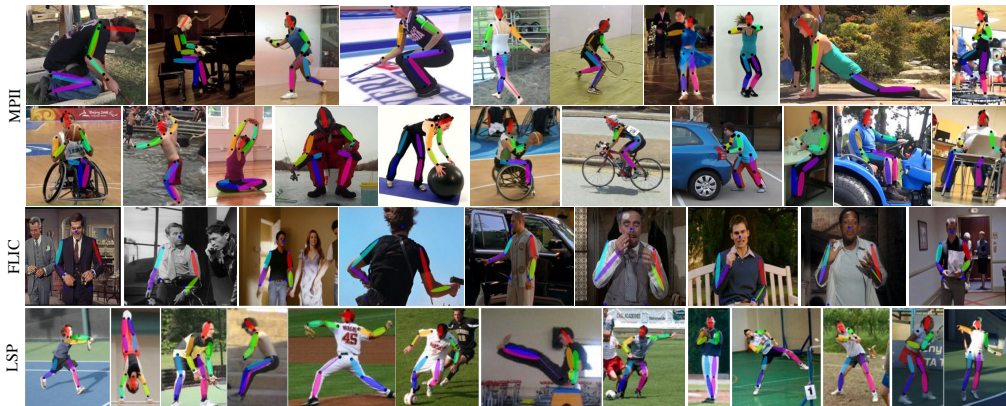


Example from Toshev and Szegedy 2014 (4.4K+ citations)

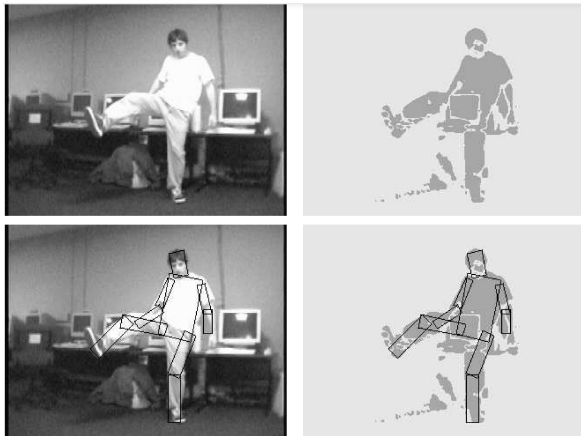




Example from Wei et al. 2016 (4.1K+ citations)



Example from Felzenszwalb and Huttenlocher 2005 (3.1K+ citations)



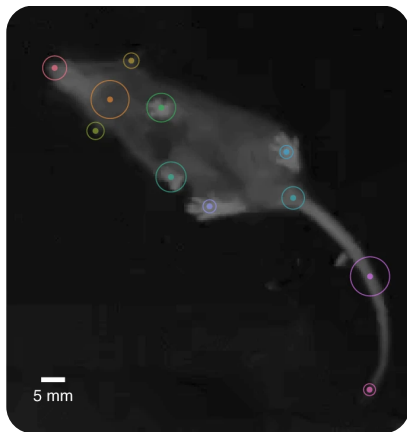


What do all of these examples have in common?

- ▶ All but one of the papers doesn't provide examples of occlusions.
- ▶ Many examples are in controlled, well lit setting.
- ▶ All but one of the papers doesn't provide examples of multi-animal settings.



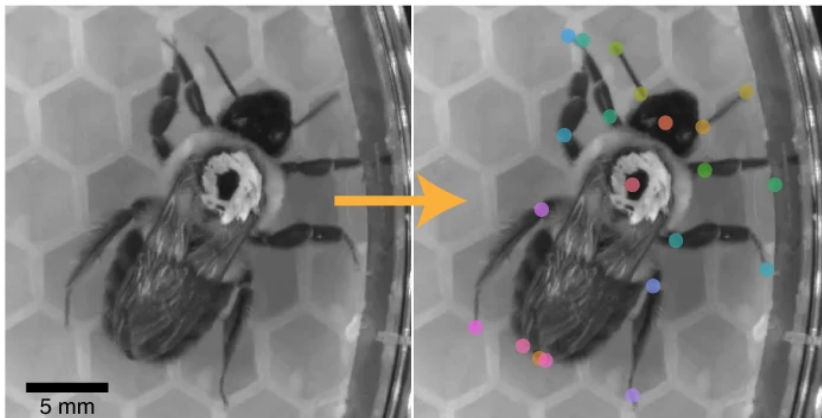
What is SLEAP?



“A deep learning system for
multi-animal pose tracking”

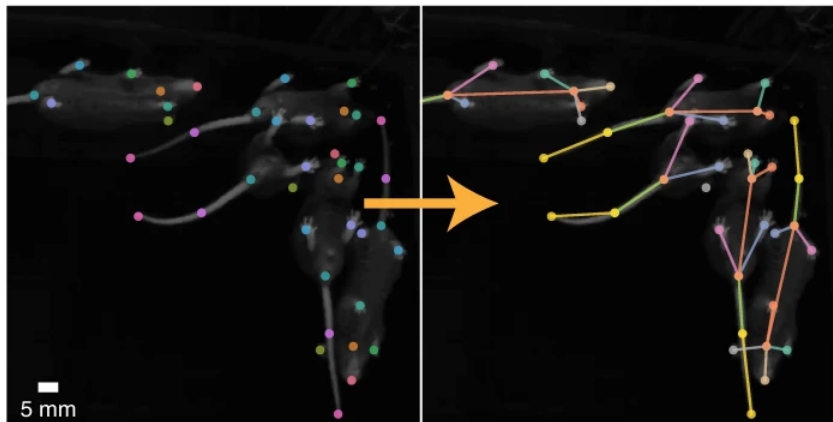
Single-animal pose estimation

Landmark localization

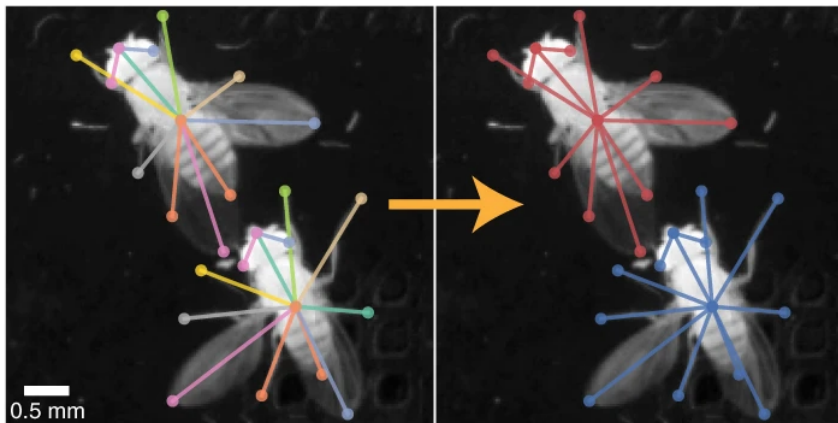


Multi-animal pose estimation

Part grouping

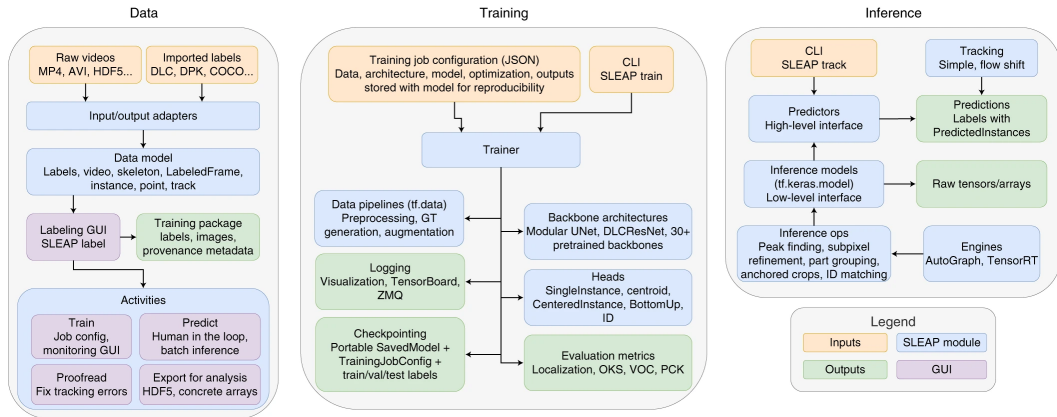


Multi-animal pose tracking Identification

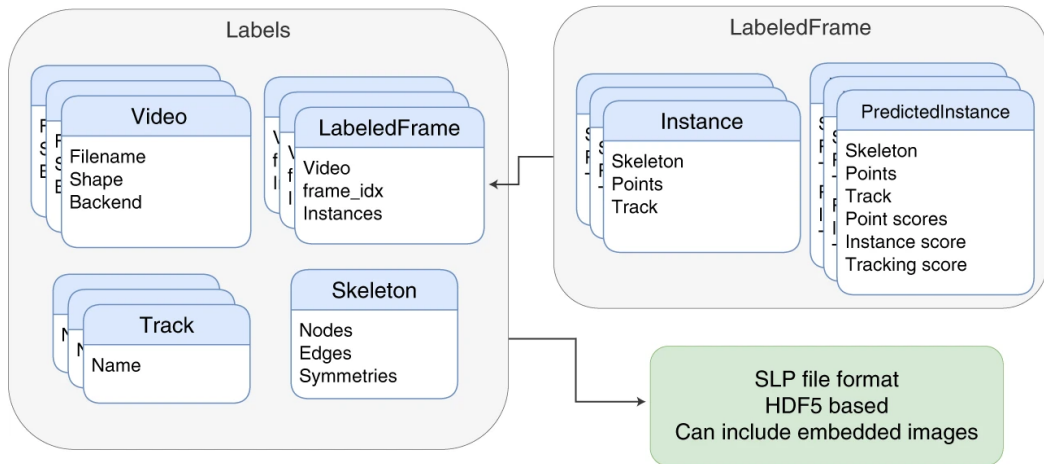


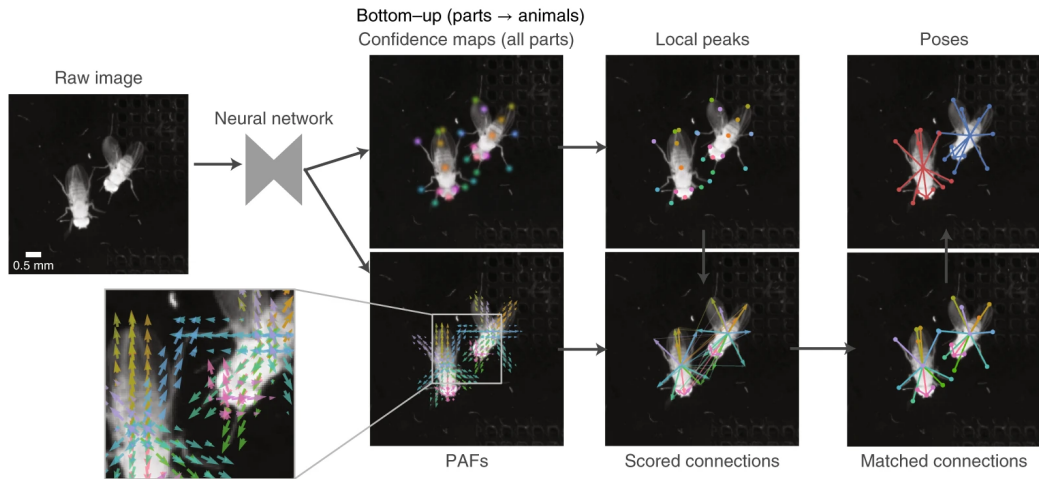


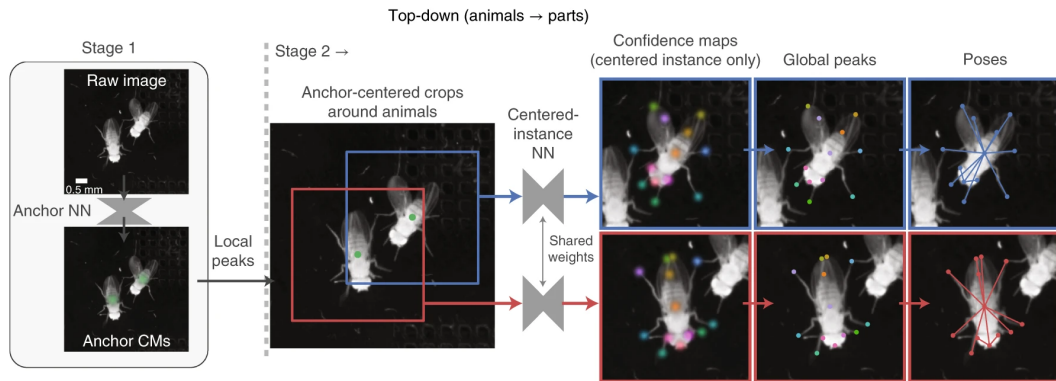
SLEAP system

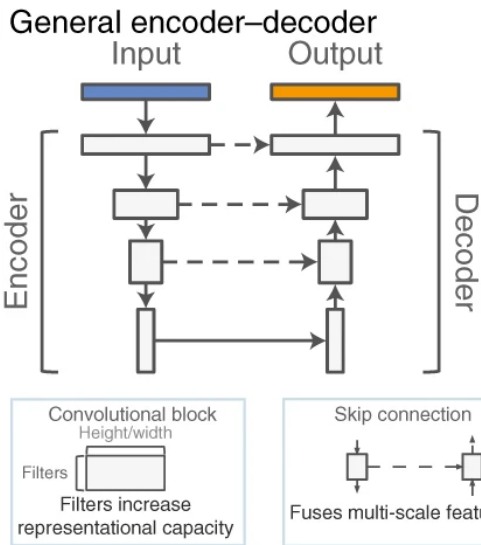


Data model

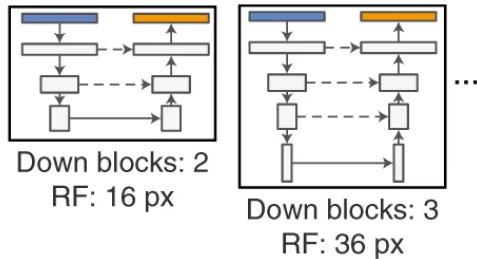








Modular UNet to control RF



Downsampling block

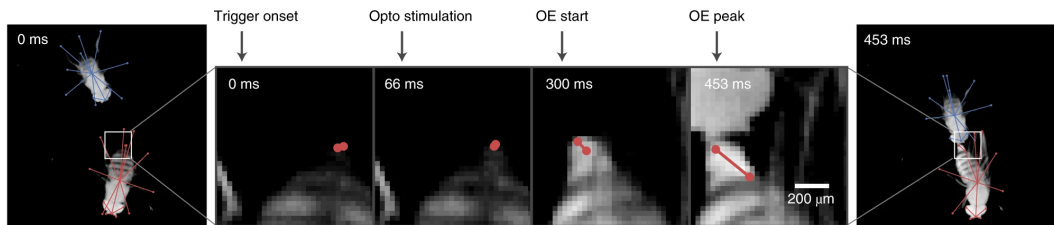


Increases RF

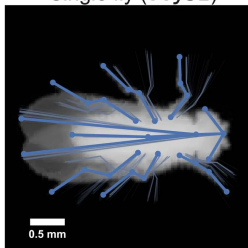
Upsampling block



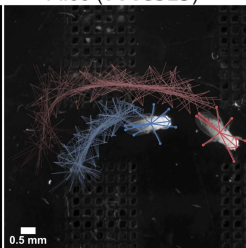
Recovers spatial resolution



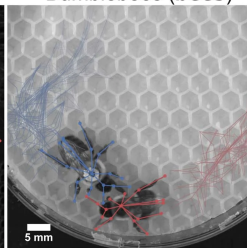
Single fly (**fly32**)



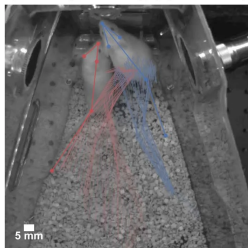
Flies (**flies13**)



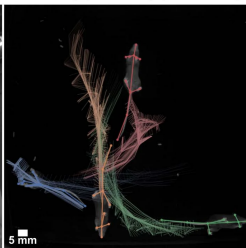
Bumblebees (**bees**)



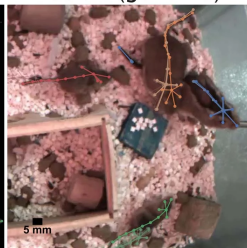
Mice (home cage) (**mice_hc**)



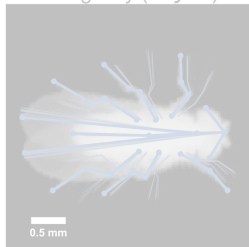
Mice (open field) (**mice_of**)



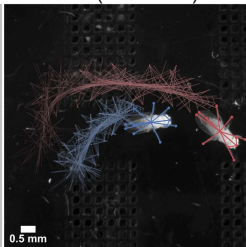
Gerbils (**gerbils**)



Single fly (**fly32**)



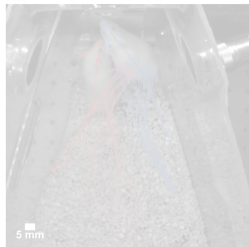
Flies (**flies13**)



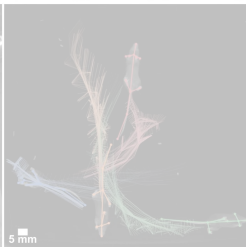
Bumblebees (**bees**)



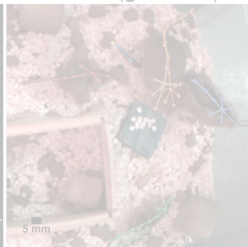
Mice (home cage) (**mice_hc**)

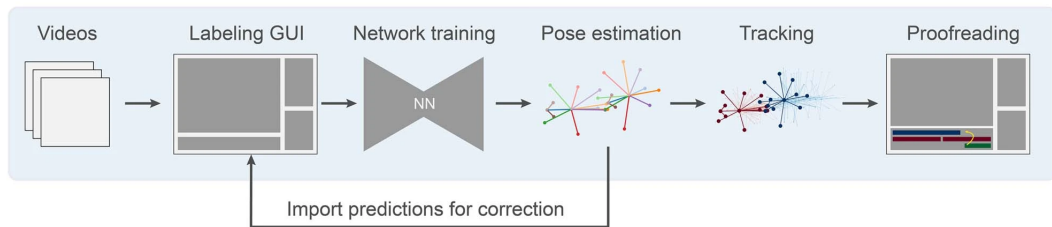


Mice (open field) (**mice_of**)



Gerbils (**gerbils**)







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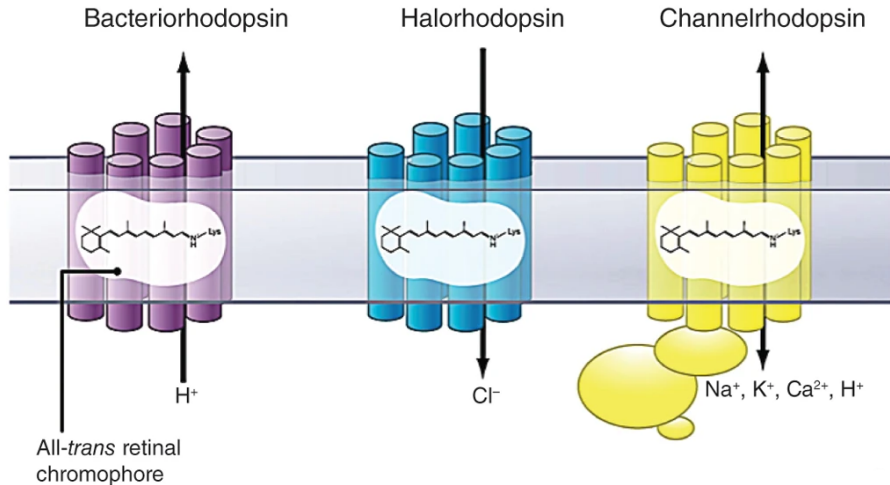


Figure 21: Adapted from Deisseroth 2015.

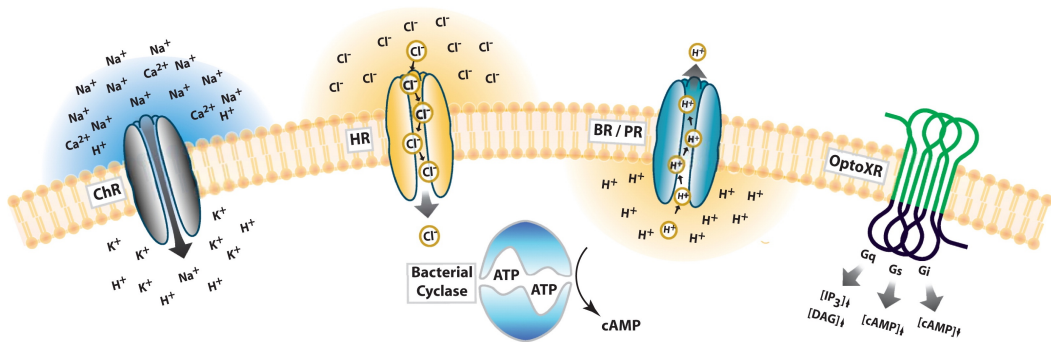
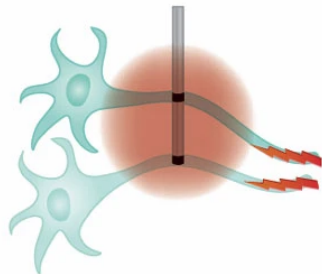
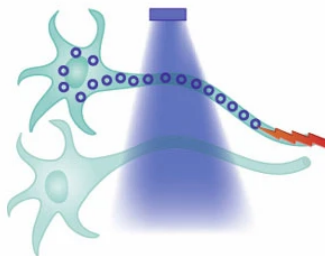


Figure 22: Adapted from Yizhar et al. 2011.

Electrical stimulation



Optogenetic excitation



Optogenetic inhibition

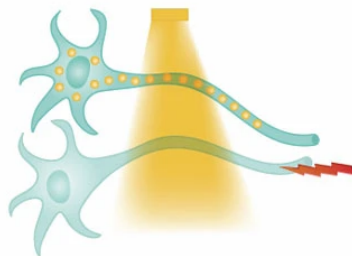


Figure 23: Adapted from Deisseroth 2011.

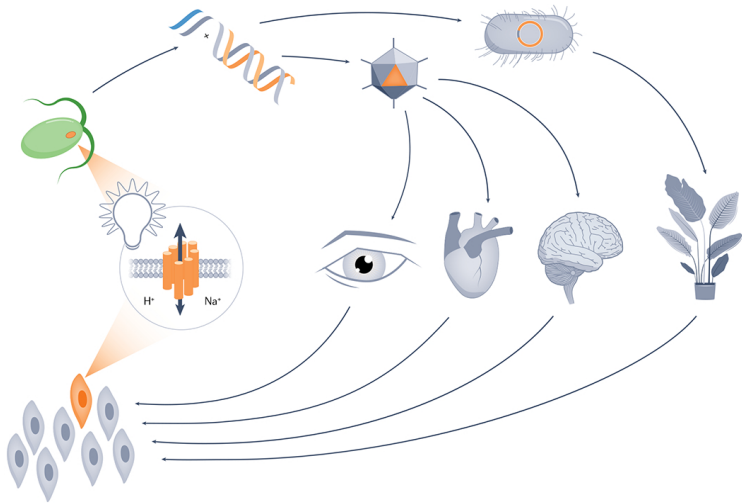


Figure 24: Adapted from Emiliani et al. 2022.

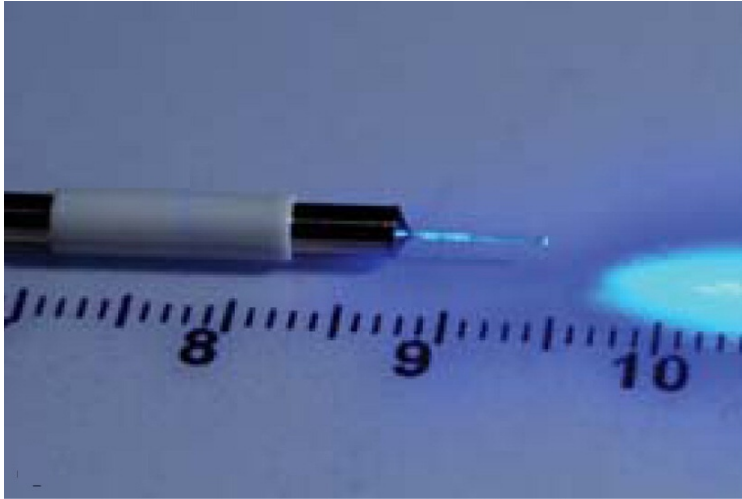


Figure 25: Adapted from Yizhar et al. 2011.

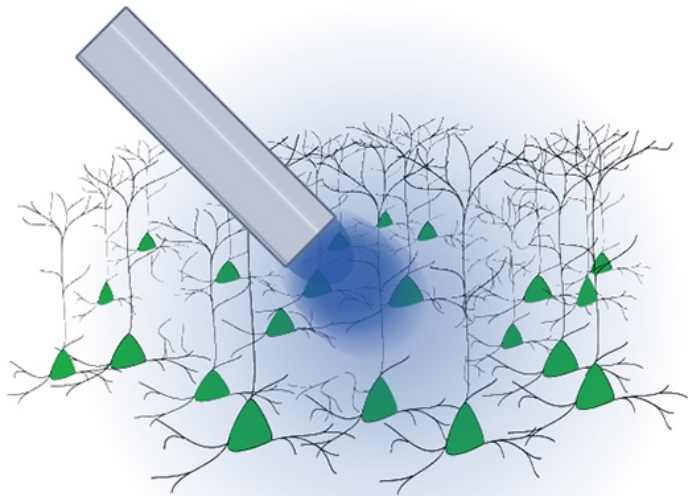


Figure 26: Adapted from Emiliani et al. 2022.

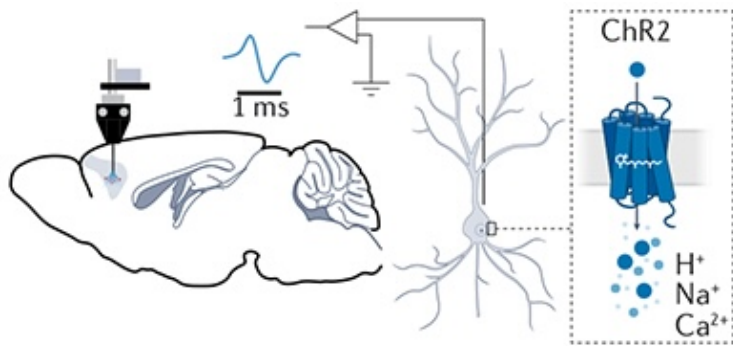


Figure 27: Adapted from Emiliani et al. 2022.

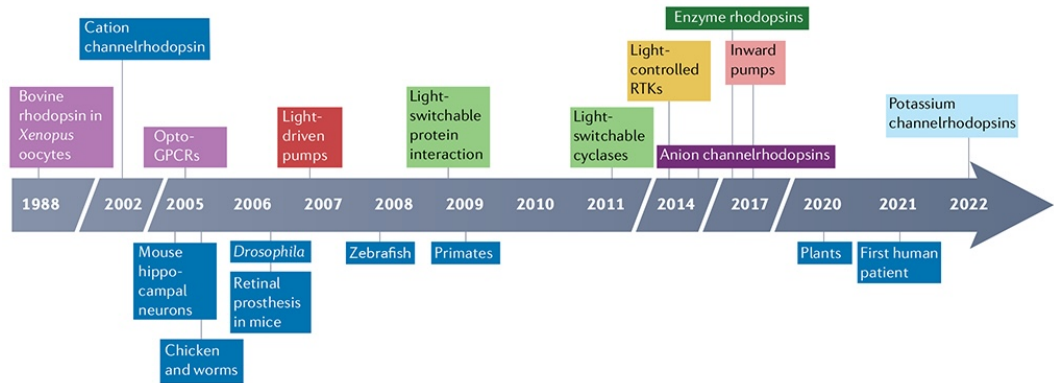


Figure 28: Adapted from Emiliani et al. 2022.

NEW

MOUSE_TRAP

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develop 2 Branches 1 Tag

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Code

This branch is 2 commits ahead of main.

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rmharpnih pre-commit fix ✓

4dcbd44 · 7 hours ago 15 Commits

.github/workflows	pre-commit fix	7 hours ago
docs	update documentation	9 hours ago
src/mouse_trap	update documentation	9 hours ago
tests	unit testing	10 hours ago
.gitattributes	Normalize line endings	last week
.gitignore	Updates	yesterday
.pre-commit-config.yaml	pre-commit fix	7 hours ago
DEVELOPMENT.md	setup type annotations	yesterday
LICENSE	Initial MOUSE_TRAP develop commit	last week
README.md	Update README	8 hours ago
pyproject.toml	pre-commit fix	7 hours ago

README License

MODULAR UNSUPERVISED POSE ESTIMATION AND TRACKING
WITH BEHAVIORAL ANNOTATION AND PREDICTION
(MOUSE-TRAP)

About

A pipeline for unsupervised quantification of social behavior.

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Contributors 2

rmharp Riley Harper

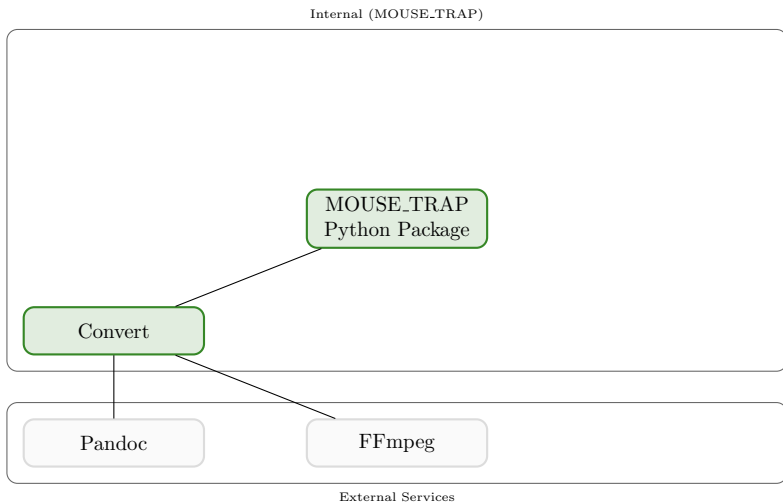
rmharpnih

Languages

Python 100.0%

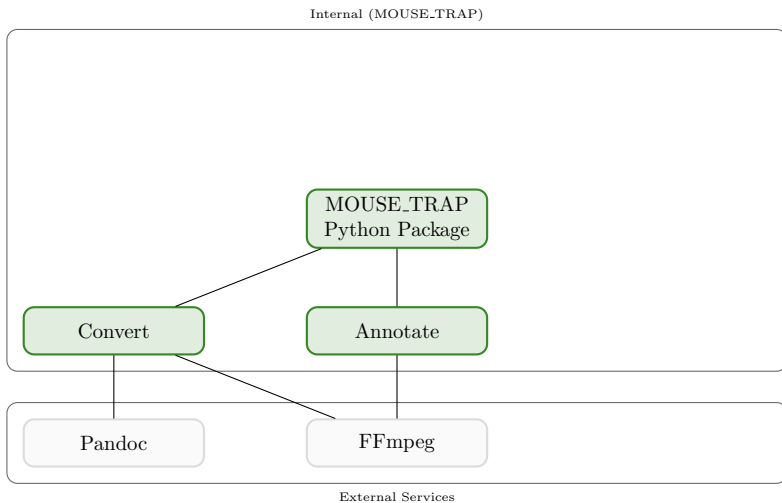


Architecture



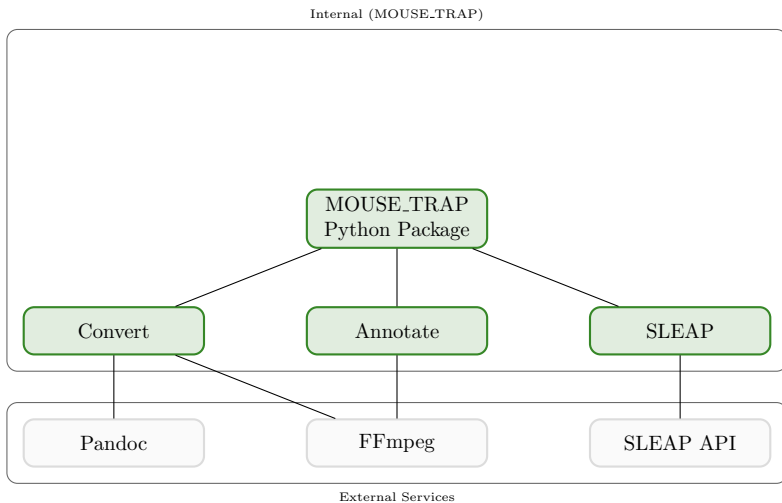


Architecture



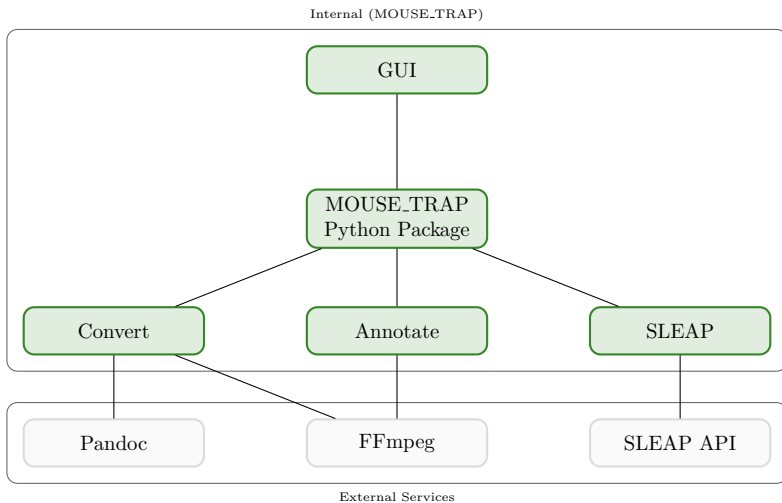


Architecture





Architecture





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Refined Objective

Convert raw video of freely moving animals into accurate keypoint-based skeletons, so that social interactions and behaviors can be quantified automatically.



Pose Estimation



Figure 30. Dynamic pose graph whose structure changes over time.



Identifying Frames of Interest

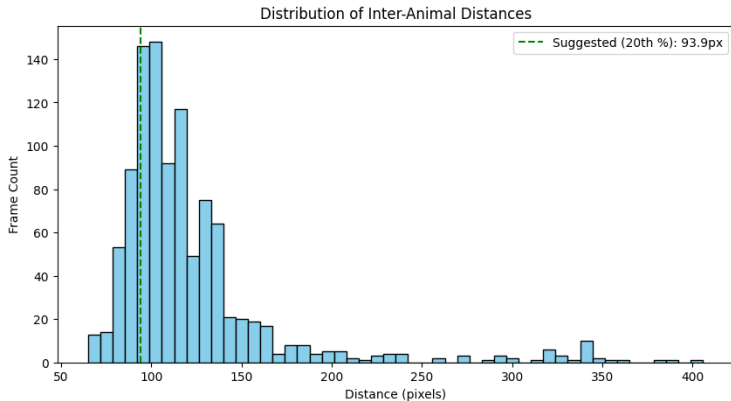


Figure 31: Right skewed unimodal distribution, with mode around 105 pixels.

Social Pose Examples

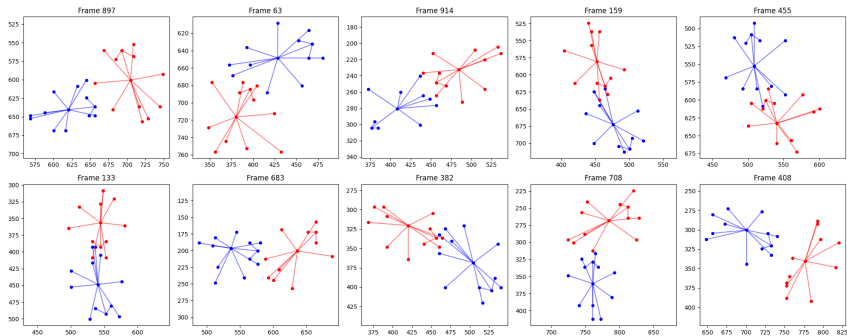


Figure 32: Ten randomly sampled social pose examples.



Where are the animals?

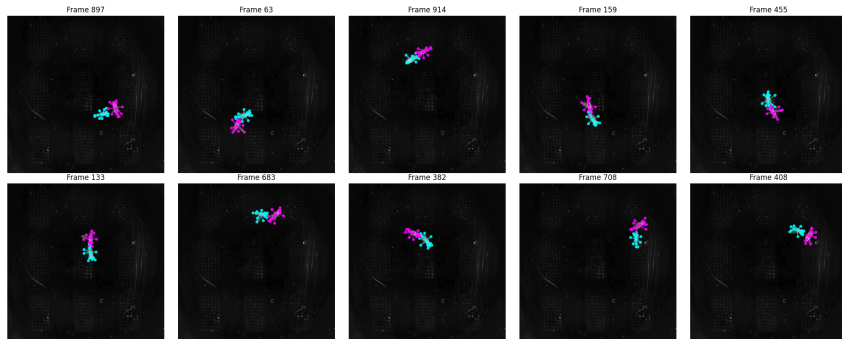


Figure 33: Investigating where they are in the cage, since some animals may prefer hugging wall or staying in corner.



What are the animals doing?

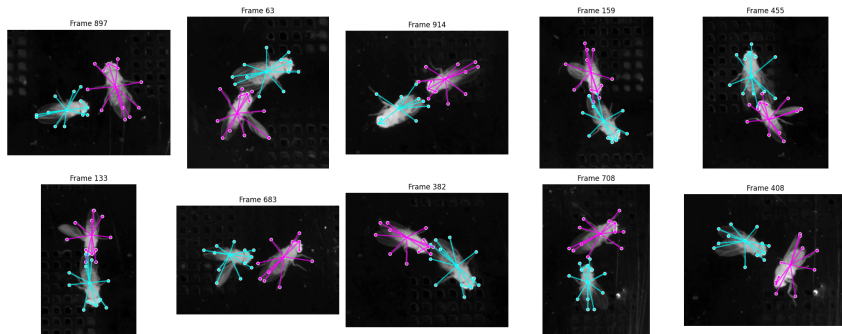


Figure 34: Investigating what they are doing with other animal.

Average Topological Pose Graph (Centered on Thorax)
Computed from 206 frames

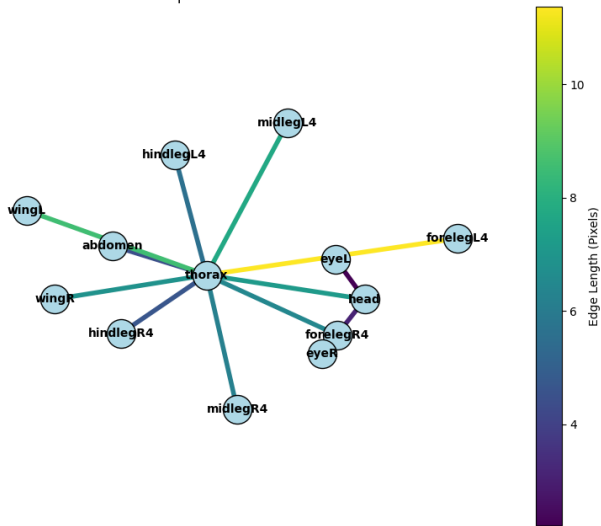


Figure 35: Average topological pose in frames identified as social interactions.

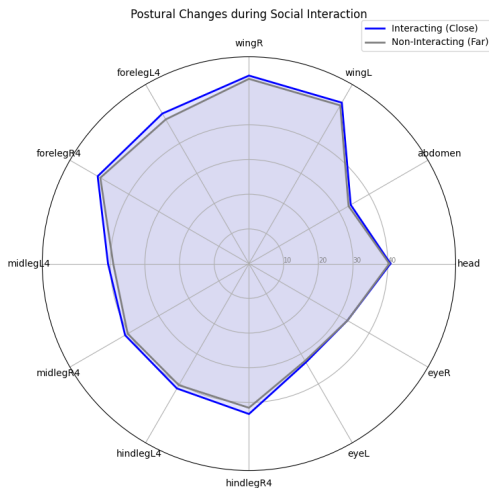


Figure 36: Comparing average topological pose in frames identified as social interactions with frames identified not to be social interactions

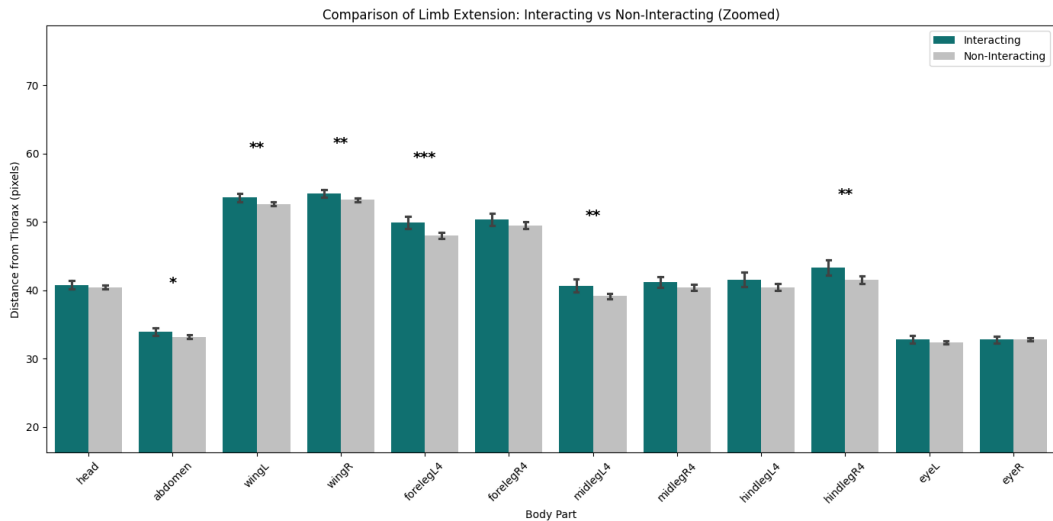


Figure 37: Asterisks mark body parts with significant differences between conditions using a Welch's two-sample t-test.



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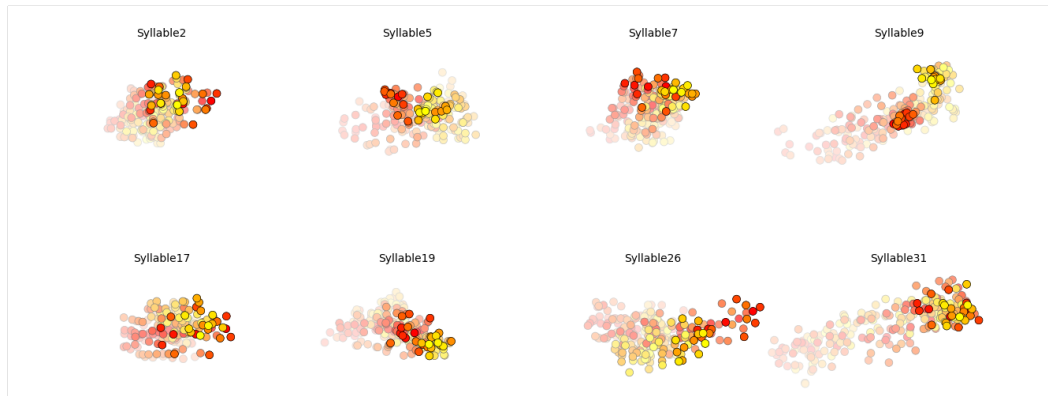


Figure 38: Identified 100 unsupervised behavior syllables, with 8 randomly selected syllables shown above.

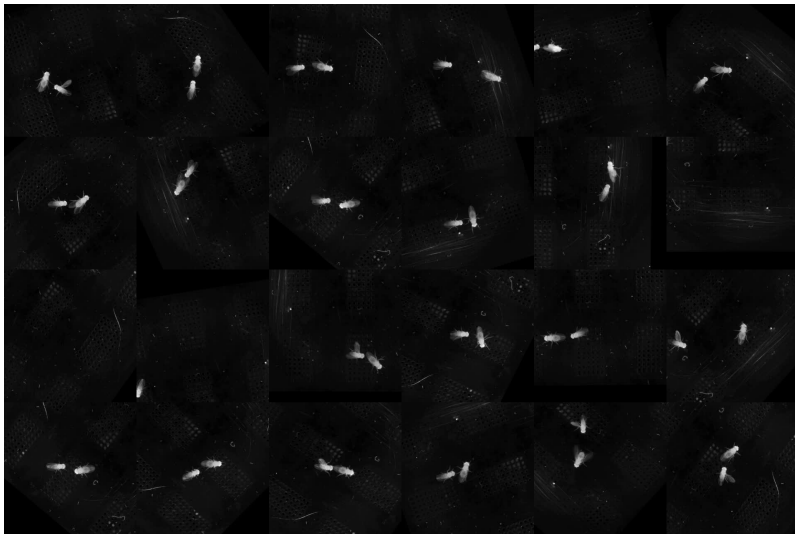


Figure 39: Investigate if these behavior syllables proved meaningful, with Syllable 63 shown above.

Fig. 1. Courtship of *D. melanogaster*. A virgin male-female pair was placed in a plastic mating cell, as implied by the border of each of these seven images. The courtship sequence proceeded in the order indicated by (A) to (G) (see text for further details). In (A), the male (large arrow) is orienting toward the female; (B) he taps the female by reaching out toward her abdomen with both of his forelegs (small arrows); (C) the male follows the female, who has almost finished flicking her wings at him, this being one of the mild rejection responses exhibited by virgin females (12); (D) the following male is also extending one of his wings, in preparation for (E) vibrating the wing to produce the courtship song; (F) the following male is also licking the female's external genitalia by extending his proboscis (small arrowhead); and (G) a copulating male-female pair. These fruit-fly adults are about 2 to 3 mm in length.

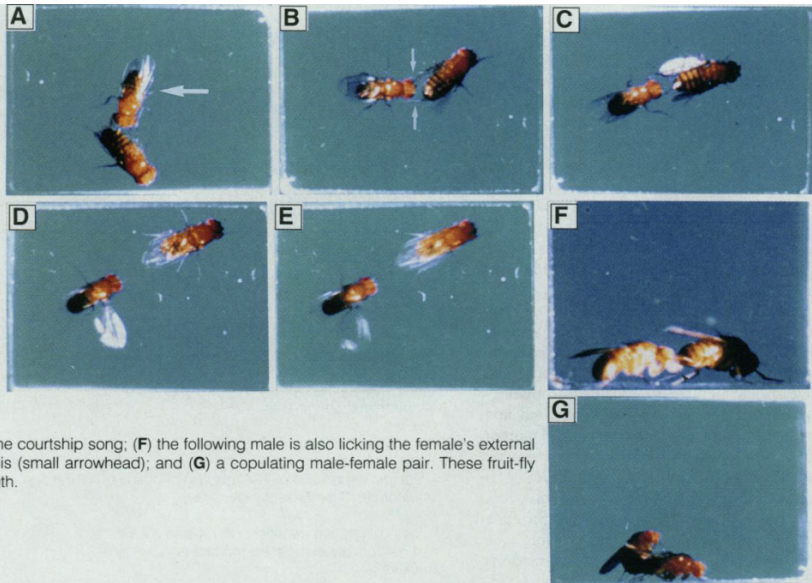


Figure 40: Adapted from Hall 1994.

Thank you for your attention!
Questions?



Works Cited



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